THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET A

Chapter 4: **Exploring Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Expand $(3x+4)^2$: 1

(a) $9x^2+12x+16$ **(b)** $9x^2+24x-16$ **(c)** $9x^2+16$ **(d)** $9x^2+24x+16$

2. Expand $(7x+1)^2$: 1

(a) $49x^2+14x-1$ **(b)** $49x^2+7x+1$ **(c)** $49x^2+1$ **(d)** $49x^2+14x+1$

3. The value of 43^2 using a suitable identity is: 1

(a) 1839 **(b)** 1849 **(c)** 1869 **(d)** 1949

4. The expression $9x^2+24x+16$ can be factorised as: 1

(a) $(3x-4)^2$ **(b)** $(3x+4)^2$ **(c)** $(3x+8)^2$ **(d)** $(9x+4)^2$

5. $(x+3)(x+4)$ equals: 1

(a) $x^2+7x-12$ **(b)** x^2-x+12 **(c)** $x^2+12x+7$ **(d)** $x^2+7x+12$

6. Factorise $x^2+11x+30$: 1

(a) $(x-5)(x+6)$ **(b)** $(x+5)(x+6)$ **(c)** $(x-6)(x+5)$ **(d)** $(x-6)(x-5)$

7. Factorise $x^2-13x+42$: 1

(a) $(x-6)(x+7)$ **(b)** $(x+6)(x+7)$ **(c)** $(x-7)(x-6)$ **(d)** $(x-7)(x+6)$

8. $(x+2)^3$ equals: 1

(a) x^3+8 **(b)** x^3+6x^2+6x+8 **(c)** x^3+2x^2+4x+8 **(d)** $x^3+6x^2+12x+8$

9. If $x+y+z=0$, then $x^3+y^3+z^3$ equals: 1

(a) $3xyz$ **(b)** $x+y+z$ **(c)** xyz **(d)** 0

10. Śhrīdharāchārya's method gives $55^2=(55+5)(55-5)+5^2=$ 1

(a) 3050 **(b)** 3000 **(c)** 2925 **(d)** 3025

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Expand $(7x+4y)^2$ using a suitable identity. 2

12. Expand $\left(\frac{5p}{2}+\frac{3q}{2}\right)^2$ using a suitable identity. 2

13. Expand $(3s-4t)^2$ using a suitable identity. 2

14. Expand $\left(x+\frac{y}{2}\right)^2$ using a suitable identity. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Factorise $9x^2 + 12xy + 6xz + 4y^2 + 4yz + z^2$. 3

16. Expand $(p + 3q + 7r)^2$. 3

17. Expand $(3x - 2y + 4z)^2$. 3

OR

Expand $(2x + 3)^3$ using $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$. 3

18. Expand $(3m - 2n)^3$. 3

OR

Factorise $8n^3 - 60n^2m + 150nm^2 - 125m^3$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Expand $(5x + 2y)^2$ and find 43^2 using the identity.
(ii) Show that $(a + b)^2 > a^2 + b^2$ when $ab > 0$ and give an example of a, b with $(a + b)^2 < a^2 + b^2$. 5

OR

Case study: Three consecutive square numbers are $(n - 1)^2$, n^2, $(n + 1)^2$. 5
(i) Add the smallest and largest and subtract twice the middle one for $n = 6$.
(ii) Prove the result is always 2.
(iii) Does the pattern hold for consecutive even-square numbers $(n - 2)^2, n^2, (n + 2)^2$? Find (smallest + largest) − 2 × middle.

20. (i) Factorise $9a^2 + b^2 + 4c^2 - 6ab + 12ac - 4bc$. 5
(ii) Factorise $16s^2 + 25t^2 - 40st$.

OR

Case study: Saira arranges 1 square of side x, 8 rectangular strips of $x \times 1$ and 15 unit squares into a rectangle. A rectangular pool has breadth 4 m less than its length and area 96 m². 5
(i) Find Saira's rectangle's length and breadth.
(ii) Form and solve the equation for the pool.
(iii) Find the pool's perimeter.

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND